

Incident report analysis

Summary	Earlier, our network services stopped responding and our internal network was compromised for two hours. Investigations indicate that an ICMP flood occurred since there was a flood of ICMP pings sent to our network. The incident management team acted quickly to restore critical network services by blocking all incoming ICMP packets.
Identify	When analyzing the cause of this issue we realized that our firewall has been unconfigured which allows any amount of packets to be sent to our network. Any IP address could also get through without any verification. This is the root cause and the main vulnerability that the malicious attacker used to overwhelm our network with the ICMP flood.
Protect	The network security team has now configured our firewall to prevent future attacks. A limit has been put on the rate of incoming ICMP packets which will maintain the flow of traffic within our network.
Detect	To detect future incidents so that they can be responded to immediately, the network security team has implemented source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets. Network monitoring software and an IDS/IPS system have also been implemented to monitor and detect suspicious behavior on the network. These new systems will work together to lock down on any threat actors lurking on our network and we will be alerted immediately when they are found.
Respond	The team has been tasked to keep up with regularly scheduled firewall maintenance and configuration checks to change and adapt our firewall configurations. Firewall patches will also be regularly updated in order to have the most secure firewall possible.



Recover	Our network was recovered relatively quickly during this attack due to immediate action and from blocking all incoming ICMP packets. All packets from the queue were cleared and all network systems were reset to restore critical network services. This process will always be used to respond to similar incidents in order to get operations back to their intended state.

Reflections/Notes: